

Mamba-based Deep Reference Frame Generation for Inter Prediction Enhancement in NNVC

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Abstract—Efficient inter prediction is fundamental to video coding, as it reduces temporal redundancy by leveraging information from neighboring frames. The quality of these reference frames critically impacts prediction accuracy. While numerous neural network-based methods have been proposed to enhance reference frames, most rely on convolutional architectures that excel at modeling local spatial features but struggle with complex motion due to their limited receptive fields. Improving their performance often demands higher model complexity, resulting in a suboptimal trade-off between performance and computational complexity. To address these limitations, we present the first exploration of a Mamba-based network that precisely captures spatiotemporal information with linear computational cost for inter prediction enhancement. Specifically, a Mamba-based video frame interpolation model is employed to synthesize a virtual reference frame using a pair of previously reconstructed frames selected from the decoded picture buffer as input. The synthesized frame is then inserted into the reference picture list and utilized during motion estimation and compensation. Moreover, we propose the QP-Progress training strategy to better align the interpolation network with the encoder's operating conditions and improve robustness to compression artifacts, without sacrificing motion modeling accuracy. This strategy progressively incorporates higher-QP compressed frames during training while retaining low-QP samples to maintain the motion modeling capacity acquired in the early training stages. Experimental results show that our method achieves coding efficiency gains of 4.46%/4.12%/4.20% for Y/U/V components under random access configuration against NNVC-12.0 anchor. Compared with other existing approaches, our method achieves superior performance gain with significantly lower complexity.

Index Terms—Neural network-based video coding, Mamba, inter prediction.

I. INTRODUCTION

Video coding technologies are essential for managing the explosive growth of video data traffic, driven by the widespread application of multimedia applications such as streaming services, video conferencing, and virtual reality. Traditional video coding standards, including Advanced Video Coding (AVC) [1], High Efficiency Video Coding (HEVC) [2], and the most recent Versatile Video Coding (VVC) [3], have achieved remarkable success by employing sophisticated hand-crafted algorithms and carefully engineered coding tools. However, these approaches are increasingly constrained by their limited ability to fully capture the complex spatiotemporal correlations inherent in video content, posing challenges to further performance improvements.

Following the finalization of the VVC standard, the Joint Video Experts Team (JVET) established Ad-Hoc Group 11 (AhG11) to explore neural network-based video coding (NNVC) [4]. Since then, AhG11 has conducted extensive research on integrating neural networks into traditional video coding frameworks, with a particular focus on coding efficiency, model complexity, and training strategies. While numerous proposals have explored neural network-based intra prediction, loop filtering, post-processing and super-resolution [5], researchers have made attempts to develop deep reference frame (DRF) generation methods aimed at further reducing temporal redundancy in video coding. For instance, Guo et al. [6] proposed a deep learning-based approach that synthesizes an additional reference frame by interpolating two previously reconstructed frames. When integrated into the reference picture list (RPL), the generated frame leads to significant performance improvements in HM-16.20 encoder. Jia et al. [7] introduced a reference frame generation framework that employs both interpolation and extrapolation networks to enhance bi-directional and uni-directional inter prediction in VTM-15.0, respectively. Their approaches were both implemented on the traditional encoder platform and demonstrated notable coding gains. Bao et al. [8] proposed a deep reference frame generation method within NNVC-5.0, demonstrating good compatibility between the DRF technique and existing neural network-based tools in NNVC framework. This method was subsequently implemented in NNVC-12.0 [9] software by Ding et al. [10]. While these methods have achieved performance gains, they still struggle to balance coding efficiency with computational complexity. This limitation is largely due to their use of convolutional neural networks (CNNs), which are well-suited for capturing local spatial features but whose small receptive fields limit their ability to capture complex motion and temporal dependencies.

Recently, State Space Models (SSMs) such as Mamba [11] have emerged as promising alternatives to traditional CNNs. These models combine linear computational complexity with the ability to capture global interactions, making them particularly suitable for video frame interpolation (VFI) tasks. Zhang et al. [12] were the first to explore the application of Mamba in VFI. They introduced the Mixed-SSM Block (MSB), which interleaves tokens from adjacent frames and applies multi-directional SSM operations to capture rich spatiotemporal dependencies. Jeong et al. [13] proposed an improved Mamba-based architecture named LC-Mamba, which explicitly incorporates local and continuous modeling capabilities. Their

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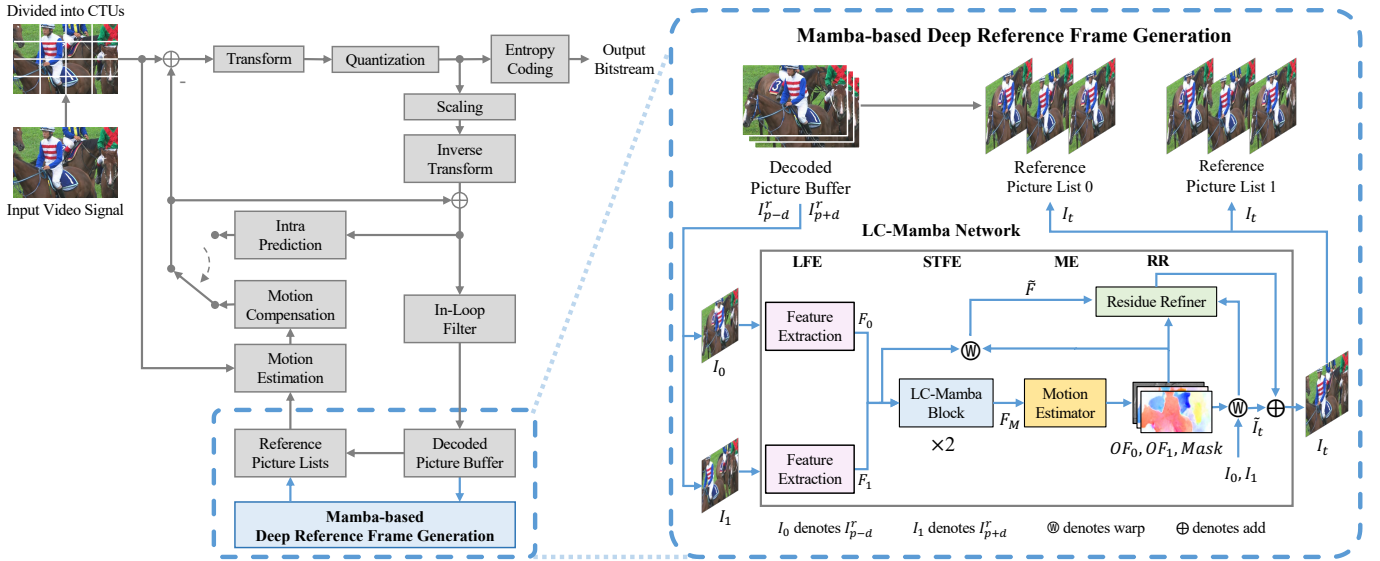


Fig. 1. The overall framework of the Mamba-based deep reference frame generation method in NNVC.

design addresses two key limitations of earlier SSM-based methods: insufficient fine-grained spatial modeling and poor temporal consistency. The LC-Mamba model achieves an effective balance between model representational capacity and computational efficiency.

In this paper, we present the first attempt to enhance inter prediction using a Mamba-based frame interpolation network. Specifically, LC-Mamba model is employed to synthesize a virtual frame by taking frames from the decoded picture buffer (DPB) as input. This synthesized frame is then inserted into RPL and used as an additional reference during motion estimation and compensation. In addition, we propose a QP-Progress training strategy that introduces higher-QP samples gradually during training while retaining lower-QP ones. This enables the model to more effectively handle diverse compression artifacts in reconstructed frames caused by encoding, while consistently preserving accurate motion modeling ability learned in the early stages, which is essential for high-quality frame interpolation. Experimental results demonstrate that our method delivers superior performance gains while operating at substantially lower computational complexity compared with existing approaches.

II. METHODOLOGY

A. Deep Reference Frame Generation

Under NNVC random access (RA) configuration, video sequences are encoded using a hierarchical group-of-pictures (GOP) structure, where each GOP consists of 32 frames distributed across 6 temporal layers. Bi-directional inter prediction is performed by referencing previously reconstructed frames from lower temporal layers, which are stored in the DPB. To enhance this process, we propose a Mamba-based deep reference frame generation method that synthesizes an additional reference frame closely approximating the current frame by leveraging temporal continuity. The overall framework of the proposed method is illustrated in Figure 1.

Specifically, for a given frame I_p at picture order count (POC) p , the closest pair of previously reconstructed frames I_{p-d}^r and I_{p+d}^r with a symmetric temporal distance d ($d \in \{1, 2, 4\}$) are selected from the DPB. These two-sided reference frames are then fed into VFI network to generate the synthesized frame:

$$I_t = \Phi(I_{p-d}^r, I_{p+d}^r), \quad (1)$$

where $\Phi(\cdot)$ denotes VFI network, and r indicates reconstructed frames.

The generated frame I_t is assigned the same POC as I_p and inserted into both reference picture lists (RPL0 and RPL1) at the second position, effectively serving as an additional two-sided reference frame during motion estimation and compensation. To ensure quality and efficiency, the deep reference frame is generated only at temporal layers 3, 4, and 5, where temporal distances between two input frames are shorter and frame similarity is higher.

B. Network Architecture

Due to the success of LC-Mamba [13] in VFI task, we adopt its architecture in our framework. As illustrated in Figure 1, it consists of four primary modules: the low-level feature extractor (LFE), the spatiotemporal feature extractor (STFE), the motion estimator (ME), and the residual refiner (RR).

Low-level Feature Extractor: The LFE is designed to capture multi-scale spatial representations from raw input frames I_0 and I_1 . It comprises a shallow convolutional encoder with hierarchical downsampling, allowing effective extraction of texture and edge information at varying resolutions. These low-level features F_0 and F_1 , obtained from:

$$F_0, F_1 = LFE(I_0, I_1), \quad (2)$$

serve as the foundation for subsequent temporal modeling.

Spatiotemporal Feature Extractor: The STFE processes the low-level features F_0 and F_1 extracted by the LFE to model dynamic dependencies across spatial and temporal

dimensions. It incorporates LC-Mamba blocks equipped with Hilbert-curve-based 3D window scanning [13]. This design allows the network to effectively capture local motion patterns while preserving temporal continuity across adjacent windows. In addition, channel attention blocks [13] are integrated to enhance the representational capacity of the extracted features, making them well-suited for subsequent processing stages such as flow estimation and adaptive fusion. The resulting spatiotemporal features F_M are obtained via the STFE as:

$$F_M = STFE(F_0, F_1). \quad (3)$$

Motion Estimator: The ME predicts bidirectional optical flow fields OF_0 and OF_1 , as well as a blending mask $Mask$ from the spatiotemporal features F_M . These flow fields are used to warp the input frames toward a target time step, thereby achieving temporal alignment for subsequent interpolation. The blending mask adaptively modulates the contributions of the warped frames to effectively address occlusions and motion uncertainty. This module plays a crucial role in producing a temporally coherent initial interpolation $\tilde{I}_t$:

$$\begin{aligned} OF_0, OF_1, Mask &= ME(F_M), \\ \tilde{I}_0 &= \overleftarrow{W}(I_0, OF_0), \\ \tilde{I}_1 &= \overleftarrow{W}(I_1, OF_1), \\ \tilde{I}_t &= Mask \odot \tilde{I}_0 + (1 - Mask) \odot \tilde{I}_1, \end{aligned} \quad (4)$$

where $\overleftarrow{W}$ denotes backward warping, and $\odot$ represents the Hadamard product. F_0 and F_1 are also warped using OF_0 and OF_1 to align them with the target frame:

$$\tilde{F}_0 = \overleftarrow{W}(F_0, OF_0), \tilde{F}_1 = \overleftarrow{W}(F_1, OF_1). \quad (5)$$

Residual Refiner: The RR enhances the initial interpolated frame $\tilde{I}_t$ by recovering high-frequency details and correcting residual spatial misalignments. It takes the optical flow fields OF_0 and OF_1 , blending mask $Mask$, intermediate interpolation result $\tilde{I}_t$ and warped low-level features $\tilde{F}_0$ and $\tilde{F}_1$ as input, and predicts a residual image that is added back to produce the final output I_t . This module is implemented as a lightweight convolutional network to ensure high visual fidelity while keeping computational overhead low. The final interpolated frame I_t is obtained by adding the predicted residual to $\tilde{I}_t$:

$$I_t = \tilde{I}_t + RR(OF_0, OF_1, Mask, \tilde{I}_t, \tilde{F}_0, \tilde{F}_1). \quad (6)$$

C. QP-Progress Training Strategy

As deep reference frame generation network used in the encoder takes reconstructed frames as input, the model is required to possess two key capabilities: accurate motion modeling to capture temporal dependencies between frames, and artifact suppression to mitigate compression distortions in reconstructed inputs. Training the model on a dataset uniformly mixed with samples compressed at various QP levels, as done in [7] and [10], was found to be suboptimal according to our experiments. Compressed samples at different QP levels contribute differently to model learning: low-QP (low distortion) samples are more effective for learning

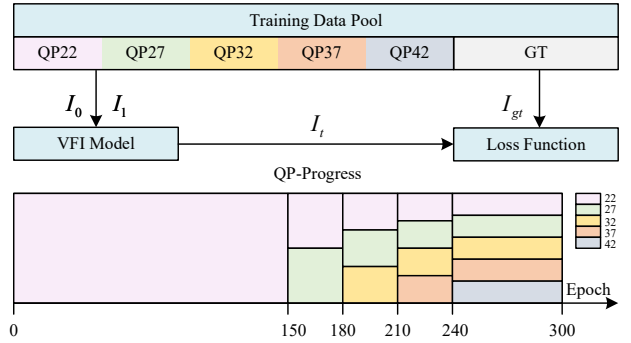


Fig. 2. The illustration of the proposed QP-Progress training strategy.

accurate motion modeling, while high-QP (high distortion) samples are more important for improving the model's robustness to compression artifacts. This difference in utility makes it challenging to optimize the model effectively when all QP levels are mixed uniformly from the beginning stage of training. To address this issue, we adopt a staged training process that gradually increases QP levels during training. In the initial training phase, the model is exposed to low-QP compressed frames, which contain slight compression artifacts. This allows the model to concentrate on learning accurate motion modeling, the core challenge in frame interpolation. In subsequent stages, higher-QP compressed frames are progressively incorporated while low-QP samples are still retained. These more distorted samples encourage the model to refine its predictions on low-quality reconstructed inputs, enhancing its ability to suppress compression artifacts while retaining the motion fidelity learned earlier. As a result, the model becomes capable of handling the full range of reconstruction qualities well encountered in practical coding scenarios.

As illustrated in Figure 2, this training strategy provides a smooth transition from low to high distortion without discarding low-QP samples. The model is trained for 300 epochs. Initially, only low-QP samples (QP=22) are used. At epoch 150, medium-quality frames (QP=27) are added to the training set. This is followed by the progressive inclusion of higher-QP samples: QP=32 at epoch 180, QP=37 at epoch 210, and finally QP=42 at epoch 240. In addition, similar to the principles of curriculum learning proposed in [14], this staged exposure facilitates smoother model optimization and leads to more stable training as shown in Figure 3 in Section III-C.

III. EXPERIMENTS

A. Training

Following the data generation strategy used by [7] [10], we construct our training dataset by compressing the Vimeo-90K triplet [15] subset using the All-Intra configuration of NNVC-12.0 [9] at five QP levels: 22, 27, 32, 37, and 42. This results in a compressed frame interpolation dataset that reflects a range of practical coding distortions. We adopt the QP-Progress strategy to train the model, using a batch size of 16 and the AdamW optimizer [16] with parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 1×10^{-4} . The learning rate is linearly increased to 2×10^{-4} over the first 2,000 warm-up

TABLE I
PERFORMANCE OF THE PROPOSED METHOD COMPARED WITH NNVC-12.0 UNDER RA CONFIGURATION

Class	Sequence	BD-rate					
		Y-PSNR	U-PSNR	V-PSNR	Y-MSIM	U-MSIM	V-MSIM
Class A1	Tango2	-5.50%	-4.95%	-5.32%	-5.82%	-5.15%	-5.67%
	FoodMarket4	-1.85%	-1.63%	-1.66%	-2.09%	-1.93%	-1.89%
	Campfire	-0.19%	-0.21%	-0.43%	-0.26%	-0.24%	-0.32%
	Average	-2.51%	-2.26%	-2.47%	-2.72%	-2.44%	-2.63%
Class A2	CatRobot	-5.53%	-5.70%	-5.63%	-5.66%	-5.48%	-5.25%
	DaylightRoad2	-6.85%	-5.75%	-5.83%	-6.68%	-6.25%	-6.12%
	ParkRunning3	-1.50%	-1.37%	-1.20%	-2.92%	-2.71%	-2.22%
	Average	-4.63%	-4.27%	-4.22%	-5.09%	-4.82%	-4.53%
Class B	MarketPlace	-3.72%	-3.23%	-4.14%	-4.14%	-3.73%	-4.32%
	RitualDance	-2.94%	-3.16%	-2.80%	-3.33%	-3.28%	-2.84%
	Cactus	-5.20%	-5.37%	-5.56%	-5.87%	-5.51%	-5.68%
	BasketballDrive	-3.58%	-3.15%	-3.23%	-4.32%	-3.96%	-3.79%
	BQTerrace	-3.38%	-2.77%	-2.98%	-3.29%	-3.26%	-3.27%
	Average	-3.76%	-3.54%	-3.74%	-4.19%	-3.95%	-3.98%
Class C	BasketballDrill	-4.35%	-3.23%	-2.82%	-6.53%	-3.73%	-4.35%
	BQMall	-7.64%	-7.38%	-7.40%	-9.22%	-8.82%	-8.10%
	PartyScene	-5.08%	-3.97%	-4.75%	-4.99%	-3.82%	-4.81%
	RaceHorses	-3.99%	-3.81%	-4.13%	-5.42%	-4.61%	-4.46%
	Average	-5.27%	-4.59%	-4.78%	-6.54%	-5.24%	-5.43%
Class D	BasketballPass	-5.77%	-6.11%	-5.68%	-7.14%	-6.91%	-5.83%
	BQSquare	-7.72%	-7.66%	-6.91%	-6.34%	-6.34%	-6.88%
	BlowingBubbles	-4.46%	-2.98%	-3.61%	-4.05%	-3.40%	-3.79%
	RaceHorses	-5.51%	-5.84%	-5.75%	-6.45%	-6.10%	-5.51%
	Average	-5.87%	-5.65%	-5.49%	-5.99%	-5.69%	-5.50%
Average		-4.46%	-4.12%	-4.20%	-4.97%	-4.49%	-4.48%

TABLE II
PERFORMANCE AND COMPLEXITY COMPARISON BETWEEN EXISTING METHODS AND OUR METHOD UNDER NNVC-12.0 RA CONFIGURATION

Class	BD-rate								
	Jia et al. [7] (1048 kMACs/pixel)			Ding et al. [10] (504 kMACs/pixel)			Ours (315 kMACs/pixel)		
	Y-PSNR	U-PSNR	V-PSNR	Y-PSNR	U-PSNR	V-PSNR	Y-PSNR	U-PSNR	V-PSNR
Class A1	-2.80%	-3.42%	-4.04%	-1.89%	-2.35%	-2.03%	-2.51%	-2.26%	-2.47%
Class A2	-4.41%	-5.92%	-5.46%	-3.27%	-2.33%	-1.81%	-4.63%	-4.27%	-4.22%
Class B	-3.32%	-4.02%	-4.34%	-2.45%	-1.11%	-1.47%	-3.76%	-3.54%	-3.74%
Class C	-4.91%	-6.66%	-7.65%	-3.65%	-1.67%	-2.52%	-5.27%	-4.59%	-4.78%
Class D	-6.14%	-6.90%	-8.16%	-4.41%	0.22%	-0.56%	-5.87%	-5.65%	-5.49%
Average	-4.34%	-5.39%	-5.97%	-3.16%	-1.34%	-1.64%	-4.46%	-4.12%	-4.20%

steps, then gradually decayed to 2×10^{-5} via cosine annealing over 962k steps, corresponding to 300 training epochs. The data augmentation and training loss follow those used in [13].

B. Experimental Results and Analysis

Following JVET NNVC common test conditions (CTC) [17], we evaluate our approach on NNVC-12.0 under RA configuration. BD-rate is used for evaluation of coding performance, and results are calculated using five QP values: 22, 27, 32, 37, 42. As shown in Table I, the average PSNR BD-rate and MSIM BD-rate metrics are -4.46%/-4.12%/-4.20% and -4.97%/-4.49%/-4.48% for Y, U, V components, respectively.

We compare our method with two representative approaches in terms of both coding efficiency and computational complexity. The method proposed in [7] was originally implemented on VTM-15.0, for a fair comparison, we re-implemented it within the NNVC-12.0, including retraining the model on the dataset compressed by NNVC-12.0. The results of [10] are directly taken from its corresponding JVET proposal document.

As shown in Table II, in terms of complexity, [7] and [10] incur 1048 kMACs/pixel and 504 kMACs/pixel, respectively, whereas our method incurs only 315 kMACs/pixel. Despite its most lightweight design, the proposed method achieves significantly better coding efficiency than [10] and even outperforms [7] on Y component, while operating at only about one-third the complexity of [7]. These results demonstrate that our method achieves a favorable trade-off between complexity and performance, which can be attributed to Mamba architecture's linear computational cost and its strong capacity for modeling inter-frame dependencies, as well as the QP-Progress training strategy that balances motion modeling accuracy and robustness to compression artifacts well.

C. Ablation Studies

We conduct an ablation study comparing the proposed QP-Progress training strategy with QP-Mixed used in [7] [10], where all quantization levels are randomly sampled from the beginning of training. Both models are trained for 300 epochs

TABLE III
ABLATION STUDY ON TRAINING STRATEGY

Class	PSNR BD-rate Over NNVC-12.0					
	(a) QP-Mixed			(b) QP-Progress		
	Y	U	V	Y	U	V
Class B	-3.38%	-3.40%	-3.54%	-3.76%	-3.54%	-3.74%
Class C	-4.83%	-4.41%	-4.82%	-5.27%	-4.59%	-4.78%
Class D	-5.17%	-4.96%	-5.02%	-5.87%	-5.65%	-5.49%
Average	-4.38%	-4.19%	-4.39%	-4.87%	-4.51%	-4.60%

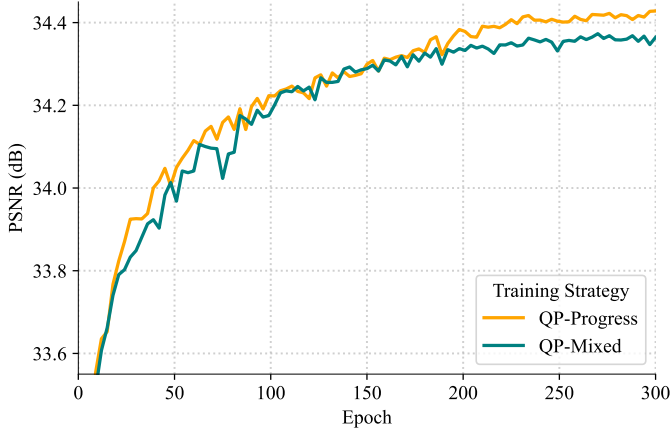


Fig. 3. Interpolation quality on compressed (QP22-42) Vimeo-90K validation set under different training strategies.

under identical settings, except for the training strategy. As shown in Figure 3, after epoch 180, the performance gap between two training strategies becomes more apparent. This divergence stems from their distinct exposure schedules to high-QP training samples. QP-Progress gradually introduces high-QP samples only after the model has already learned accurate motion representations from high quality (low-QP) frames. This staged progression allows the model to adapt to compression artifacts without disrupting the previously acquired motion modeling capability. In contrast, QP-Mixed is exposed to both low-QP and high-QP frames from the beginning, which causes the model to prioritize artifact suppression early on, leaving limited capacity for learning fine-grained motion features. As a result, its performance improves more slowly in the later training stages. When evaluated using PSNR BD-rate over the NNVC-12.0 anchor, the QP-Progress training strategy consistently outperforms the QP-Mixed across all test classes as shown in Table III.

IV. CONCLUSIONS

This paper presents the first integration of a Mamba-based interpolation network for inter prediction enhancement in NNVC. Leveraging the LC-Mamba architecture, the proposed method effectively captures temporal dependencies between frames while maintaining linear computational complexity, addressing the limitations of convolution-based DRF methods. Furthermore, to better align the training process with encoding scenarios involving diverse compression levels, we propose the QP-Progress strategy. It gradually incorporates higher-

QP samples while retaining lower-QP ones. This progressive training strategy improves generalization across compression levels, enhances training stability, and yields a better balance between motion modeling and artifact suppression compared to the QP-Mixed training strategy. Experimental results demonstrate that our method achieves average PSNR BD-rate of -4.46%/-4.12%/-4.20% over NNVC-12.0 anchor under RA configuration. Notably, it achieves these gains with the lowest computational complexity about 315 kMACs/pixel among comparable methods.

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